



Communicable disease

Name: _____

Class: _____

Date: _____

Time: **65 minutes**

Marks: **62 marks**

Comments:

1.

Monoclonal antibodies (mAbs) are usually made using mouse lymphocytes.

Candida albicans infection produces serious symptoms in patients with a poor immune system.

Recently scientists have produced mAbs to *Candida albicans* using human lymphocytes produced naturally after an infection.

(a) *Candida albicans* lives in the throat of infected patients.

A sample is taken from the throat of a patient with a suspected *Candida albicans* infection.

The sample is transferred onto a microscope slide.

Describe how the mAbs and a fluorescent dye could be used to see any *Candida albicans* pathogens on the slide.

(3)

In a laboratory the human lymphocyte mAbs were injected into animals infected with *Candida albicans*.

The mAbs caused increased phagocytosis of the *Candida albicans* pathogens.

Doctors intend to start a trial to give the mAbs to patients severely ill with *Candida albicans*.

(b) Explain how increased phagocytosis of the *Candida albicans* pathogen will help the patient.

(2)

- (c) It has been shown that this mAbs treatment is effective in the laboratory using both:
- infected tissue culture cells
 - infected live animals.

The mAbs treatment for *Candida albicans* is now ready for clinical trials on people.

Describe how the clinical trials should be carried out.

(6)

- (d) Scientists have also used human lymphocytes to make mAbs to other pathogens and to some types of cancer cells.

Suggest **one** reason why these new mAbs have been more successful in treating diseases in humans than mAbs made using mice.

(1)

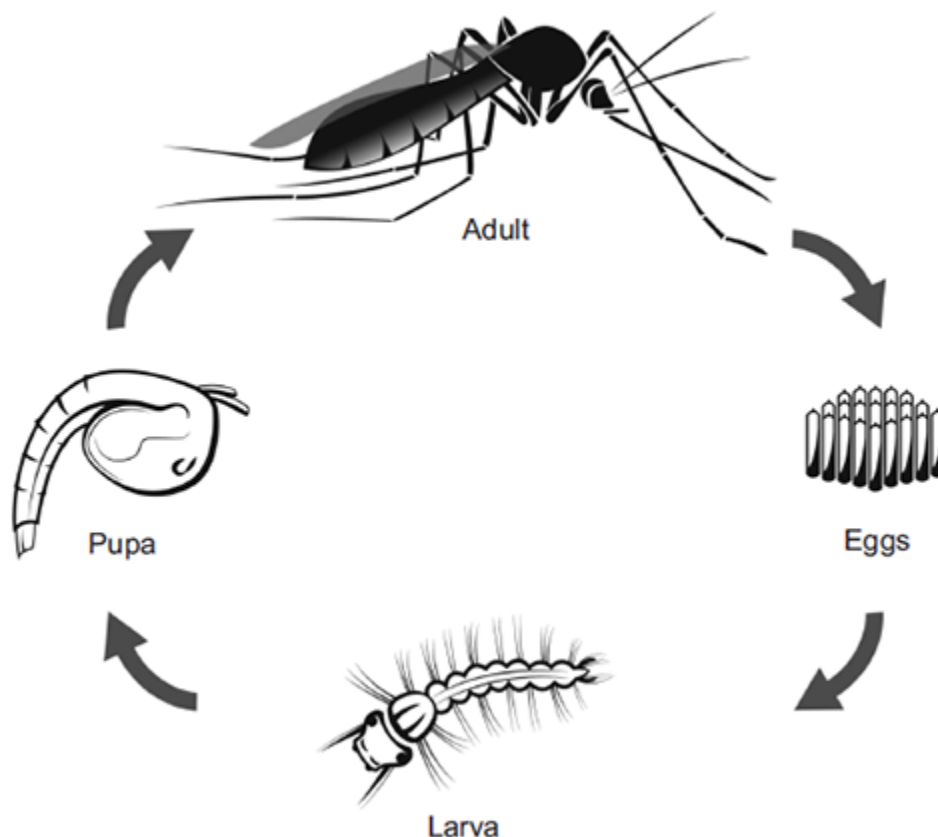
(Total 12 marks)

2.

Malaria is a disease caused by a microorganism carried by mosquitoes.

The microorganism is transferred to humans when adult female mosquitoes feed on human blood.

The figure below shows the life cycle of a mosquito.



© watcharapon/iStock

The World Health Organisation estimates that 3×10^8 people are infected with malaria every year.

Scientists estimate that malaria kills 2×10^6 people every year.

The people who are infected with malaria but do not die, may be seriously ill and need health care for the rest of their lives.

- (a) Based on the estimated figures, what percentage of people infected with malaria die from the disease?

(2)

- (b) An internet article states:
- 1 Mosquito larvae are at the start of the food chain for some fish.
 - 2 Adult mosquitoes provide food for bats and birds.
 - 3 Mosquitoes are also important in plant reproduction because they feed from flowers of crop plants.

- (i) The first sentence in the article is **not** correct.

Explain why.

(2)

- (ii) A company plans to produce genetically modified (GM) adult male mosquitoes. The GM mosquitoes will carry a gene from bacteria. The gene causes the death of offspring before they become adults.

Male mosquitoes do **not** feed on blood.

Scientists are considering releasing millions of adult male GM mosquitoes into the wild.

Do you think scientists should release millions of male GM mosquitoes into the wild?

In your answer you should give advantages and disadvantages of releasing GM mosquitoes into the wild.

(4)

(iii) Describe the process for creating a GM mosquito.

(3)

(Total 11 marks)

3.

A virus called RSV causes severe respiratory disease.

(a) Suggest **two** precautions that a person with RSV could take to reduce the spread of the virus to other people.

1. _____

2. _____

(2)

(b) One treatment for RSV uses monoclonal antibodies which can be injected into the patient.

Scientists can produce monoclonal antibodies using mice.

The first step is to inject the virus into a mouse.

Describe the remaining steps in the procedure to produce monoclonal antibodies.

(3)

- (c) Describe how injecting a monoclonal antibody for RSV helps to treat a patient suffering with the disease.

(2)

A trial was carried out to assess the effectiveness of using monoclonal antibodies to treat patients with RSV.

Some patients were given a placebo.

- (d) Why were some patients given a placebo?

(1)

A number of patients had to be admitted to hospital as they became so ill with RSV.

The results are shown in the table below.

Treatment received by patient	% of patients within each group admitted to hospital with RSV
Group A : Monoclonal antibody for RSV	4.8
Group B : Placebo	10.4

The trial involved 1 500 patients.

- Half of the patients (group **A**) were given the monoclonal antibodies.
- Half of the patients (group **B**) were given the placebo.

- (e) Calculate the total number of patients admitted to hospital with RSV during the trial.

Total number of patients admitted to hospital = _____

(2)

(f) Evaluate how well the data in the table above supports the conclusion:

‘monoclonal antibodies are more effective at treating RSV than a placebo’.

(2)

(Total 12 marks)

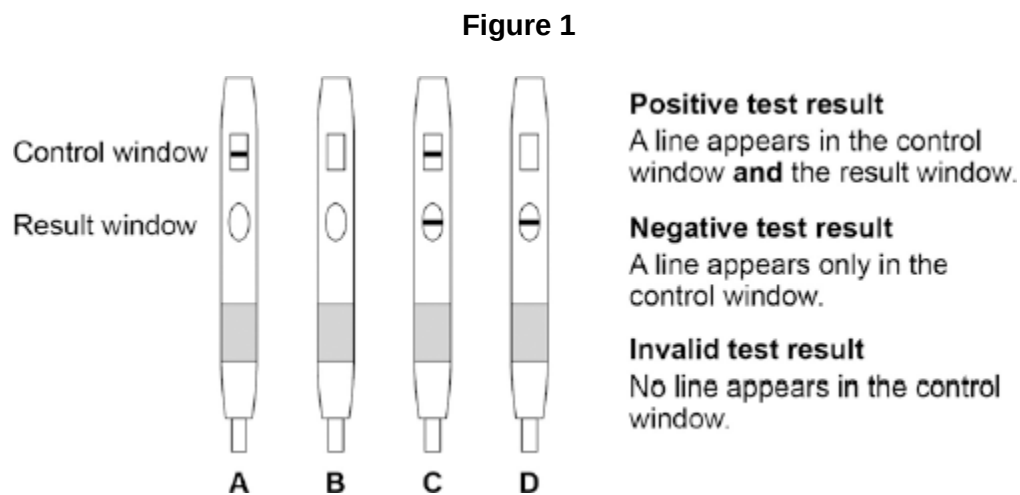
4.

Monoclonal antibodies are used to measure the levels of hormones in the blood.

Pregnant women produce the hormone HCG.

HCG is excreted in urine.

Figure 1 shows four pregnancy test strips.



(a) Which test strip shows a negative test result?

Tick **one** box.

A ☐ **B** ☐ **C** ☐ **D** ☐

(1)

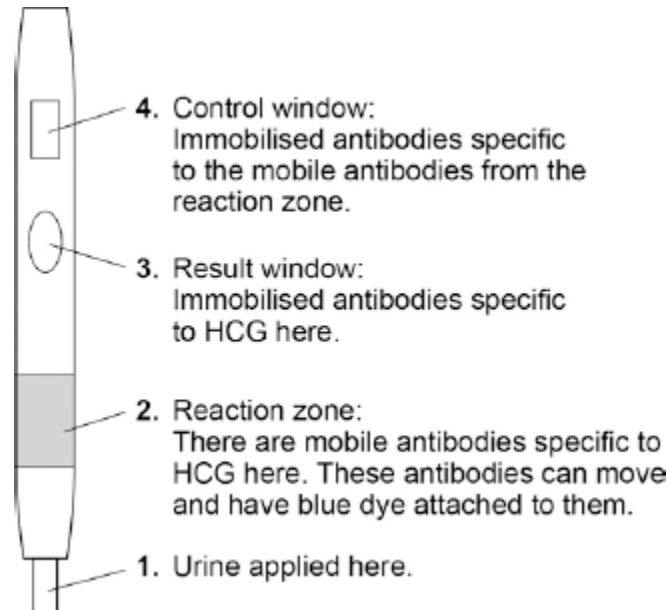
(b) Monoclonal antibodies are used for pregnancy testing.

Give **one other** use of monoclonal antibodies.

(1)

(c) **Figure 2** shows the parts of a pregnancy test strip.

Figure 2



The pregnancy test strip will show a positive test result when a woman is pregnant.

Explain how the pregnancy test strip works to show a positive result.

(6)

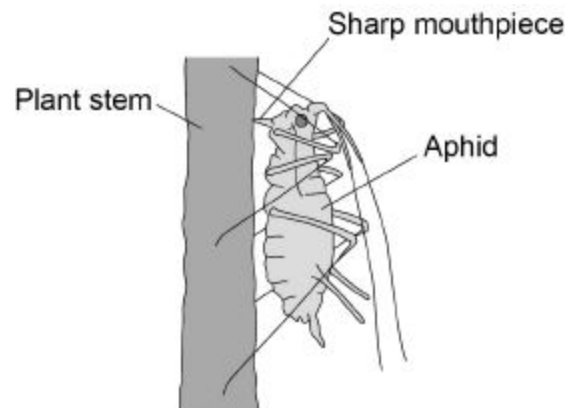
(Total 8 marks)

5.

Plants can be infected by fungi, viruses and insects.

Aphids are small insects that carry pathogens.

The diagram below shows an aphid feeding from a plant stem.



- (a) An aphid feeds by inserting its sharp mouthpiece into the stem of a plant.

Give the reason why the mouthpiece of an aphid contains a high concentration of dissolved sugars after feeding.

(1)

(b) Plants infected with aphids may show symptoms of magnesium deficiency.

Magnesium deficiency symptoms include:

- yellow leaves
- stunted growth.

Explain how a deficiency of magnesium could cause these symptoms.

(5)

- (c) A farmer thinks a potato crop is infected with potato virus Y (PVY).

The farmer obtains a monoclonal antibody test kit for PVY.

To make the monoclonal antibodies a scientist first isolates the PVY protein from the virus.

Describe how the scientist would use the protein to produce the PVY monoclonal antibody.

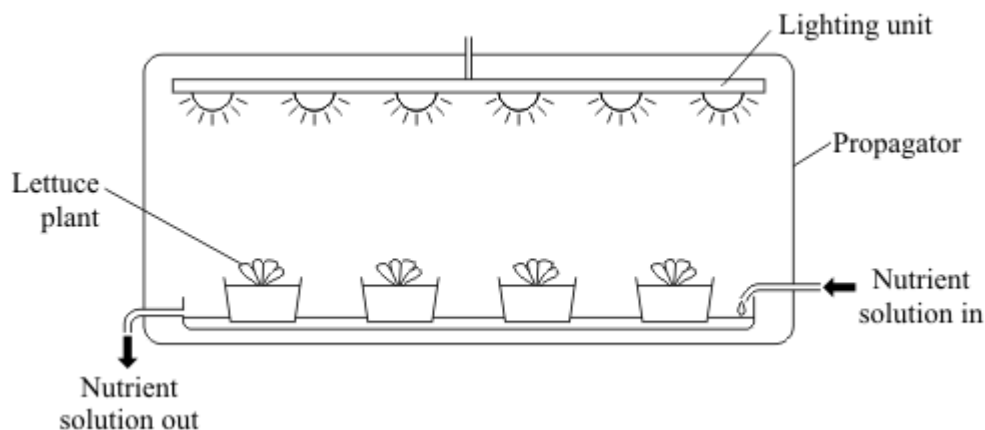
(4)

(Total 10 marks)

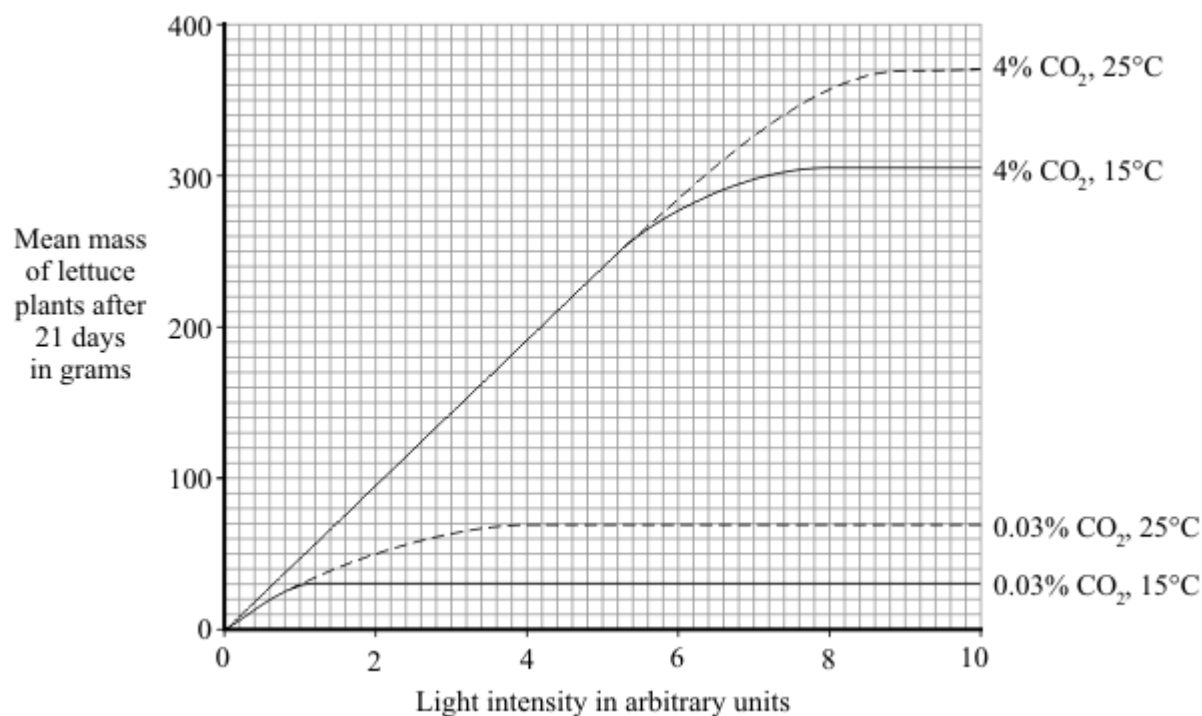
6.

Changing the conditions in which plants grow affects how fast they grow.

The diagram shows a propagator in which scientists can control temperature, light intensity and carbon dioxide concentration.



The graph shows the effects of changing the temperature, light intensity and carbon dioxide concentration on the growth of lettuce plants.



- (a) Describe and explain the effect of increasing light intensity on the mean mass of lettuce plants at 4% carbon dioxide and 15 °C.

(3)

- (b) Growers wish to make maximum profits from their lettuces.

What do they need to consider before making decisions about the growing conditions for their lettuces?

(2)

- (c) The nutrient solution contains nitrate ions and magnesium ions.

Complete the table to show the functions of these ions in plants and their deficiency symptoms.

Ion	Function in plants	Deficiency symptoms
Nitrate		
Magnesium		

(4)

(Total 9 marks)

Mark schemes

1.

- (a) bind fluorescent dye to mAbs

1

put (bound) fluorescent mAbs on the slide (and rinse off)

ignore add mAbs and dye to slide (unbound)

1

mAbs will bind to *Candida albicans* / pathogens and show up under the microscope

*allow mAbs will bind to Candida albicans / pathogens
and show up under UV (lamp)*

1

- (b) more *Candida albicans* / pathogens will be engulfed / killed by phagocytes / white blood cells

*allow Candida albicans / pathogens will be engulfed /
killed by phagocytes / white blood cells more quickly
do **not** accept white blood cells produce antibodies
do **not** accept lymphocytes engulf Candida albicans*

1

therefore less damage to cells / tissues / organs

ignore less toxin released (by Candida albicans)

1

- (c) **Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.

4–6

Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.

1–3

No relevant content

0

Indicative content

- given first to healthy volunteers
 - at (very) low dose
 - to test it is safe **or** to test for toxicity **or** to check for any side effects
- then to some patients (with the disease) **or** people with the disease
 - to test for the correct / optimum dose
 - to check for any side effects
 - to test for efficacy **or** to test if it works
 - in a double blind trial
 - where neither patients nor doctors know who has the mAbs and who has a placebo (or alternative treatment)
- reference to large trial **or** long duration **or** control variables

- (d) any **one** from:
- (the body will) not reject the mAbs
or (the body is) less likely to reject the mAbs
*do **not** accept idea of rejection of cells*
 - mouse mAbs are (more likely to be) rejected
 - the human lymphocytes have already responded to that infection / cancer cell
so they are known to work against the disease

1

[12]

2.

- (a) 0.67(%)

allow 0.6 or 0.7

allow 1 mark for evidence of $(2 \times 10^6) \div (3 \times 10^8)$

or

allow 1 mark for 0.0067 or 0.6

2

- (b) (i) idea that food chains start with plants / producers

*allow food chains do not start with animals **or** larvae are consumers*

1

idea that these make food (for other organisms in the chain)

*allow idea that plants / producers photosynthesise **or** plants / producers get energy from the sun*

*allow mosquito larvae do not make food / photosynthesise **or** mosquito larvae do not get energy from the sun*

1

- (ii) any **four** from:
- reasoned argument for **or** against release
must refer to at least one advantage and one disadvantage.
*max 3 marks for either only advantages **or** only disadvantages*

advantages:

- fewer mosquitos biting **or** spreading malaria
- fewer people get / die from malaria
allow people won't get / die from malaria
- lower medical costs (for those infected **or** for treatment) **or** less healthcare needed
- better economically for developing / tropical countries.

disadvantages:

- fewer crops reproduce
allow fewer crops pollinated
- poorer crop yield
- possible starvation (of people)
- high cost of GM production / mosquito release
- less food for bats / birds **or** bats / birds die
*allow disruption to food chain / ecosystem **or** reduction of biodiversity*
- gene could 'escape' into other wildlife / species
ignore into plants

4

- (iii) any **three** from:
- gene from bacteria cut out
allow allele for gene
 - ref to enzymes (anywhere in process)
allow at any point in process, ie in cutting or in splicing
 - (gene) transferred to chromosome of mosquito
allow DNA for chromosome
 - at an early stage of development
allow egg / embryo

3

[11]

3.

- (a) any **two** from:
- regular hand washing
or
use hand sanitiser / alcohol gel
 - cover nose / mouth when coughing / sneezing
allow wear a face mask
 - put used tissues (straight) in the bin
 - don't kiss uninfected people
allow isolate patient from others
- or**
- don't share cutlery / cups / drinks with uninfected people
- clean / disinfect / sterilise surfaces regularly
ignore responses referring to infected people

2

- (b) any **three** from:
- stimulate (mouse) lymphocytes to produce antibody
for marking points 1 and 2 lymphocyte must be used at least once
 - combine (mouse) lymphocyte with tumour cell
or
(create a) hybridoma
 - clone (hybridoma) cell
 - (hybridoma) divides rapidly **and** produces the antibody

3

- (c) any **two** from:
- (monoclonal) antibody binds to virus **or** antibody binds to antigen on surface of virus
 - (monoclonal) antibody is complementary (in shape) / specific to antigen (on surface of virus)
 - white blood cells / phagocytes kill / engulf the virus(es)

2

- (d) as a control
or
to see / compare the effects of the treatment (vs. no treatment)

1

- (e) $(4.8 + 10.4) \div 2 \div 100 \times 1500$
or
 $(4.8 \div 100 \times 750) + (10.4 \div 100 \times 750)$

1

114

an answer of 114 scores 2 marks
allow 228 for 1 mark

1

- (f) **(supports the conclusion because)**
over double the number / % of patients (in the trial) were hospitalised with the placebo
(compared to MAB)

1

(does not support the conclusion because)
no information on patients not hospitalised / still unwell at home
or
other factors may have affected those admitted to hospital
allow correct named factor e.g. age / gender / other illness
or
don't know if it was a double blind trial

1

[12]

4.

- (a) **A**

1

- (b) any **one** from:
- identify / locate specific molecules / other hormones
 - locate blood clots
 - diagnose / treat some cancers

1

- (c) (as) urine passes through reaction zone

1

HCG hormone binds to the mobile HCG antibody (in the reaction zone)

1

(passes up the stick) HCG hormone binds to the immobilised HCG antibodies in the results zone

1

(the other) antibodies which do not attach to HCG

1

bind to antibodies in control zone

1

blue dye appears in both control and results zones (to show positive result)

1

[8]

5.

- (a) (mouthpiece) has pierced / entered the phloem
or
(the aphid) has been feeding from the phloem

1

- (b) yellow leaves due to lack of chlorophyll

ignore 'chloroplasts'

ignore magnesium is needed to make chlorophyll

1

(therefore) less / no light absorbed (by chlorophyll)

1

(therefore) lower rate of / no photosynthesis

*do **not** allow 'energy is produced by photosynthesis'*

1

(therefore) plant makes less / no sugar / glucose

1

(therefore) plant converts less / no sugar / glucose into protein (for growth, so growth is stunted)

allow less glucose / sugar converted into cellulose (cell wall)

allow less energy for protein synthesis

1

(c) inject the protein / it into a mouse

1

combine lymphocytes with tumour / cancer cells to make hybridoma (cells)

ignore white blood cells

allow T or B lymphocytes

ignore tumour unqualified

1

find a hybridoma which makes a monoclonal antibody specific to PVY

1

(the scientist) clones (the hybridoma) to produce many cells (to make the antibody)

*do **not** allow cloning of original stem cells*

allow many rounds of cloning / mitosis

1

[10]

6.

(a) any **three** from:

- ((mean) mass) increases up to 7 / 8 units (of light) then levels off
- light limiting factor up to 7 / 8 units
- for photosynthesis
must be in correct context
- other factor / temperature limiting above 7 / 8 units

3

(b) any **two** from:

- cost of providing conditions / heat / light / CO₂
- effect of treatment on profit
allow too much of factor is wasteful
- relevant use of data from graph eg limiting factors
- named other factors eg fertiliser / pest control / weeds / density of planting
allow taste / appearance

2

(c) **nitrate function**

produce amino acids / proteins / enzymes

ignore DNA

*do **not** allow chlorophyll*

1

nitrate deficiency

stunted growth

allow description

ignore plant dies

1

magnesium function

produce chlorophyll

ignore chloroplasts

1

magnesium deficiency

yellow leaves / plant

ignore plant dies

1

[9]

Examiner reports

1.

Many students unnecessarily explained how monoclonal antibodies were produced for (a), and very few scored marks on this question. Many students misunderstood the context of the question, and indicated that the mAbs would be injected into the slide. Some students thought that the mAbs were made of lymphocytes. In (b), several students incorrectly stated that the mAbs themselves or lymphocytes would engulf the pathogen.

In (c), many students were able to access Level 2, however, a number of students made blanket statements about clinical trials that applied to all participants. Several students were unable to clarify the difference between patients and healthy volunteers, with some using confusing terminology such as 'healthy patients'.

2.

- (a) A little under a quarter of students gained both marks for this calculation. It was clear that standard form had confused some students, although many who gained only one mark had failed to round the calculated figure correctly, offering '0.6', rather than '0.67', '0.7' or '0.6 recurring'. An alternative route to gaining the one mark was to show the division sum required and some of those who did this multiplied their answer by 100 twice, to arrive at 67%.
- (b) (i) A significant number of students interpreted the first sentence as meaning that mosquito larvae eat fish, which was clearly incorrect. Others suggested that fish could not eat larvae as fish live in water while larvae live on land or in the air, although the habitat of larvae had not been given. Consequently few students gained both marks here. Those who did recognise the flaw in the first sentence of the article often stopped after pointing out that food chains must start with plants or producers or that the larvae are not producers, and did not go on to explain the vital role of producers in the food chain as carrying out photosynthesis or making food.
- (ii) Students often failed to use all the information in the question and consequently missed out on potential marks. Many took the short term view that the release of mosquitoes would increase food for birds or bats or allow greater crop production, whilst others suggested that these male mosquitoes would provide no threat as they do not bite humans so cannot pass on the malaria microorganism. What examiners were looking for was the awareness that scientists must be carrying out the release for some reason and that could only be the reduction of the numbers of mosquitoes in the long term, and the consequences of this on other wildlife. There was also a common misconception that because the gene was transferred from a bacterium that somehow the GM mosquitoes would infect people with the bacterium and cause disease in this way. There was a surprising amount of concern that mosquitoes may be made extinct as a result of the release. Some wrote that all the GM mosquitoes would be genetically identical and therefore could all be wiped out by a disease. Many students repeated statements from the stem of the question eg 'adult mosquitoes provide food for birds and bats' without adding any value and so did not gain credit. Consequently relatively few students scored more than two marks for this part.

- (iii) Where students wrote about genetic modification, this question was generally answered well. Unfortunately many students wrote about a different technique altogether, most commonly adult cell cloning. Students were often unclear about where the gene was being cut from (the bacteria) or did not get the idea of 'cutting out' across well enough. The third marking point was also often not given because students did not make it clear where the gene was being transferred to. Weaker students were unclear as to whether the gene was being transferred from a mosquito to a bacterium or vice versa, or described the injection of complete bacteria into the mosquito.

5.

- (a) Many students recognised the relevance of the phloem, although some of these simply stated that the phloem carries sugars, without linking this to feeding by the aphid.

Some students continued the theme from the previous question and linked 'high concentration' in the question to diffusion, osmosis or active transport.

- (b) Although many students showed some understanding of the reason for stunted growth, most did not go on to achieve three or four marks.

The lack of magnesium was often linked to chlorophyll deficiency, causing yellow leaves, although chloroplasts were sometimes confused with chlorophyll. The fact that reduced chlorophyll would lead to a reduction in light absorption was frequently omitted as students moved directly to a reduction in photosynthesis and in turn, reduced glucose production.

However a lack of glucose was only rarely linked to a shortage of protein or cellulose, but more commonly to a lack of energy which was insufficient. Many students were able to describe much of the process that happens when magnesium is not deficient but did not answer the question by stating that none of this would happen in the case of magnesium deficiency.

- (c) This was a new area of the specification and although many students knew that something had to be injected, most were confused either as to the organism receiving the injection or to the contents of the syringe.

At the next step the need for lymphocytes, cancer cells and the production of a hybridoma was not understood by many. Although many knew the term 'hybridoma' most answers described the wrong components. Specificity was only mentioned on very rare occasions, as was the need for the scientists to clone the hybridoma. Many students did know some details, but clarity of detail and explanation was frequently lacking.

6.

Part (a) discriminated between those candidates who carefully read the question and information and those who did not. Weaker candidates often compared the growth, in general terms, as shown by all four curves and were able to gain few marks, or failed to read the instruction to describe and explain. Other candidates lost marks by not quoting values, between 7 and 8 units of light, from the graph where the curve levelled off. Many candidates were aware that limiting factors were involved, but understanding was insufficiently secure to gain marks for suggesting which factors might be limiting at each stage of the curve. Even the weakest candidates usually managed to refer to photosynthesis somewhere in their answer.

Part (b) was well answered by most candidates who recognised that the gardener would need to pay for the extra light etc. However some failed to pick up the second mark due to their answer being too vague in relation to the effect of this treatment on the profit. The cost of nutrients was a popular response which gained a lot of candidates a mark. Very few candidates were able to use relevant data accurately from the graph as part of their argument.

It was hoped that the majority of candidates taking the Higher Tier paper would have scored all four marks in part (c), where straightforward recall was required. It was very clear to examiners which candidates had learnt this, quoting verbatim from the specification in their answer and consequently scoring full marks. Where candidates fell down was when their knowledge was not secure and they decided to hedge their bets and combine symptoms, often quoting yellow leaves and stunted growth as deficiency symptoms for both ions or by completely reversing the function and deficiency symptoms of the two ions.